

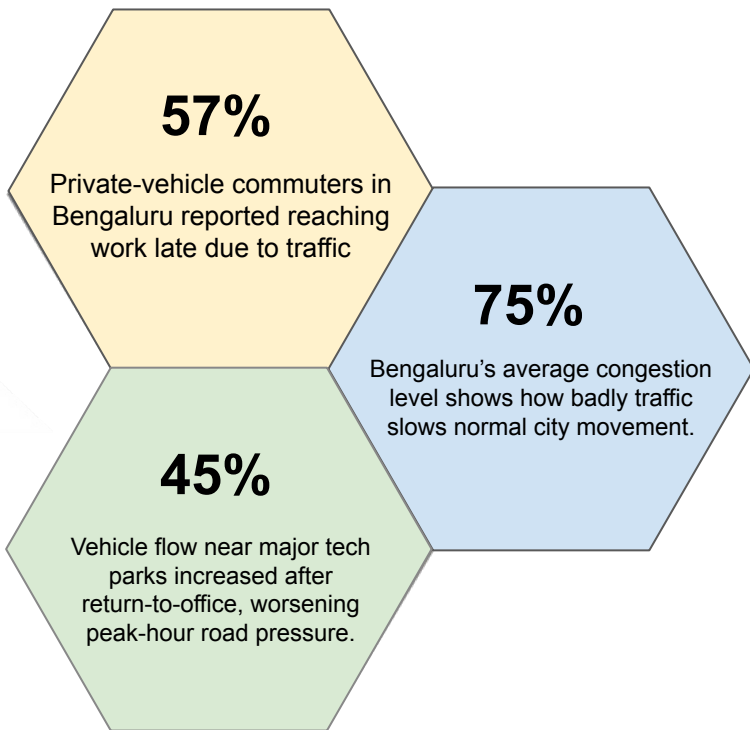


TRAFFIC DEMAND PREDICTION

Team Name: .0anamta

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Problem Overview



Problem Statement:

Bengaluru is one of India's most ambitious cities. Getting around in it shouldn't be a daily compromise.

The final goal is to build a model that predicts the demand column for each test record accurately.

Methodology

Predict demand for the unseen part of day 49. The strongest signal is the same place and time on day 48, then adjusted using the first two hours of day 49.

77,299

training rows

41,778

test rows

11 → 10

train/test columns

92.49

final selected score

70%

spatial backbone

14%

CatBoost direct

8%

LightGBM residual

6%

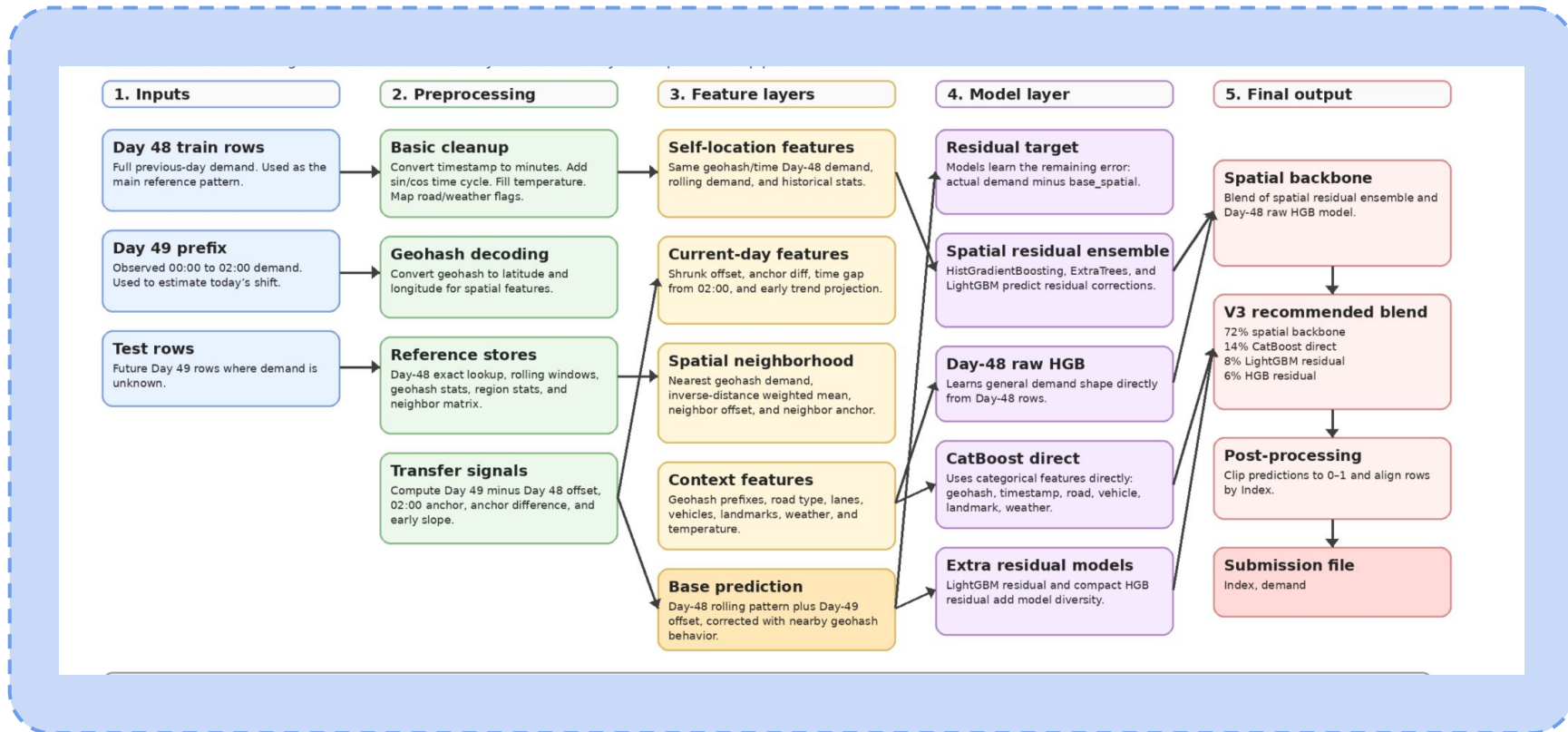
HGB residual

2%

previous best

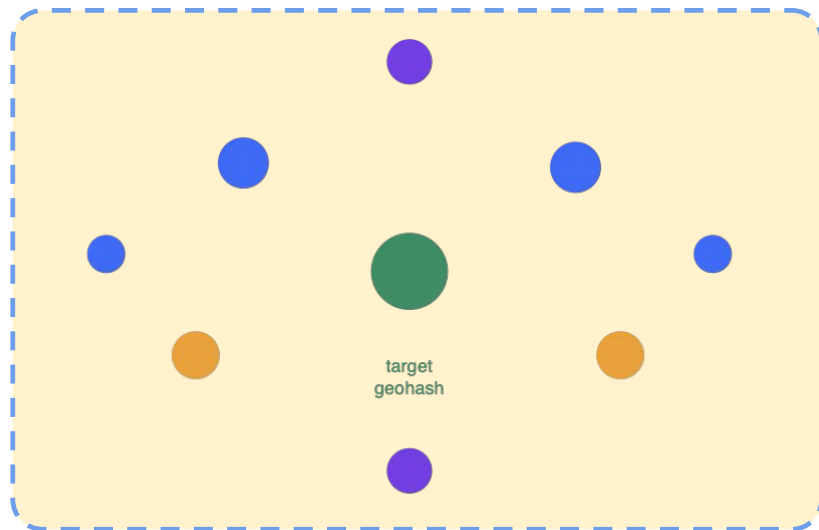
Solution Overview

Architecture:



The Spatial Neighborhood Layer

Instead of treating each geohash alone, the V3 pipeline borrows signal from nearby geohashes. This captures spillover patterns when demand shifts across adjacent cells.



Neighbor features

Nearest 8 geohashes, inverse-distance weighted demand, neighbor offsets, neighbor anchor difference, and neighbor availability.

Region features

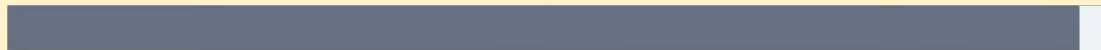
Short geohash prefixes add local area averages so the model sees both micro-neighborhood and wider-zone behavior.

Time transfer

The known 00:00–02:00 rows estimate whether each location is running hotter or colder than day 48.

Worked with multiple methodologies

Early ensemble



90.47

good baseline, but limited by per-geohash view

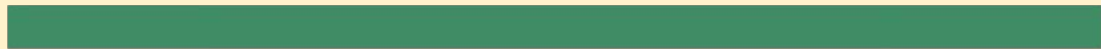
Spatial global blend



92.40

first major jump from neighbor signal

Final blend



92.49

final selected clean/global submission

About
92.5%
accuracy

Future Scope

Enhanced Model Training with Larger Datasets: Continue fine-tuning the model with diverse datasets to enhance extraction accuracy and speed.

Scalability through GPU Upgrades: Leverage more powerful GPUs to reduce training time and enable fine-tuning with larger models for better accuracy.

Appendices

GitHub Repository:

https://github.com/AnamtaRehman/flipkart_gridlock_2.0

References:

cureusjournals.com/articles/1981-traffic-congestion-prediction-using-machine-learning-algorithm

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THANK YOU